

Lecture L11_12: Transformation of Random Variables

Dhaval Patel, PhD

CSE400: Fundamentals of Probability in Computing

February 10, 2026

1 Outline

- [cite_sstart]
- **Transformation of Random Variables:** Learning techniques for $y = g(x)$ [cite: 30, 33]. [cite_sstart]
- **Function of Two Random Variables:** Joint transformations and derived distributions $f_Y(y)$ [cite: 34, 37]. [cite_sstart]
- **Illustrative Example:** Detailed derivation for the case $Z = X + Y$ [cite: 39, 40].

2 Transformation of a Single Random Variable

[cite_sstart] Given a random variable X with a known PDF $f_X(x)$ and CDF $F_X(x)$, we seek to find the PDF $f_Y(y)$ of a new random variable $Y = g(x)$ [cite: 1, 21, 23, 24].

2.1 Derivation Steps

- [cite_sstart]
- **Step 1: Find the CDF of Y** [cite: 51]
[cite_sstart] $F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$ [cite: 60, 61]
(1)
[cite_sstart]
- **Step 2: Differentiate with respect to y to find the PDF** [cite: 50, 52]
[cite_sstart] $f_Y(y) = \frac{d}{dy}[F_X(g^{-1}(y))] = f_X(g^{-1}(y)) \cdot \frac{d}{dy}g^{-1}(y)$ [cite: 52, 54]
(2)

2.2 General Formula

If $g(x)$ is a monotonic function, the PDF of Y is expressed as:

$$f_Y(y) = \frac{f_X(x)}{\left| \frac{dy}{dx} \right|} \bigg|_{x=g^{-1}(y)} \quad [\text{cite: 75, 90}] \quad (3)$$

3 Worked Example: Uniform to Sine Transformation

Problem: Let X be a uniformly distributed RV on $(-1, 1)$. *Find the PDF of $Y = \sin\left(\frac{\pi x}{2}\right)$* [cite: 80].

Solution:

1. **Given PDF of X :**

$$f_X(x) = \begin{cases} \frac{1}{2} & -1 < x < 1 \\ 0 & \text{otherwise} \end{cases} \quad [\text{cite: 82}] \quad (4)$$

[cite_start]

1. **Transformation:** $y = \sin\left(\frac{\pi x}{2}\right) \implies x = \frac{2}{\pi} \sin^{-1}(y)$ [cite: 83, 84].

[cite_start]

1. **Derivative:** $\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$ [cite: 85].

2. **Resulting PDF:**

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}} = \frac{1}{\pi \sqrt{1-y^2}}, \quad -1 < y < 1 \quad [\text{cite: 92, 93}] \quad (5)$$

4 Function of Two Random Variables: $Z = X + Y$

To find the PDF $f_Z(z)$ when $Z = X + Y$:

4.1 CDF Method

$$F_Z(z) = \Pr(X + Y \leq z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{XY}(x, y) dx dy \quad [\text{cite: 121, 123}] \quad (6)$$

4.2 Leibniz Rule for Differentiation

To differentiate the integral $G(x) = \int_{a(x)}^{b(x)} h(x, y) dy$:

$$[cite_start] \frac{d}{dx} [G(x)] = \frac{d}{dx} b(x) \cdot h(x, b(x)) - \frac{d}{dx} a(x) \cdot h(x, a(x)) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} h(x, y) dy \quad [cite: 132, 133] \quad (7)$$

4.3 PDF of Sum (Convolution)

$[cite_start]$ If X and Y are independent, the PDF of Z is the convolution of f_X and f_Y [cite: 99, 154]:

$$[cite_start] f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx \quad [cite: 154] \quad (8)$$

5 Applications of Joint Transformations

$[cite_start]$

- **Sum of Normals:** If $X \sim N(0, 1)$ and $Y \sim N(0, 1)$ are independent, then $Z = X + Y \sim N(0, 2)$ [cite: 102, 103, 177]. $[cite_start]$
- **Sum of Exponentials:** If X, Y are exponential RVs with parameter λ , find $f_Z(z)$ using the convolution integral [cite: 104, 183, 189]:

$$[cite_start] f_Z(z) = \int_0^z \lambda e^{-\lambda x} \cdot \lambda e^{-\lambda(z-x)} dx \quad [cite: 191] \quad (9)$$

$[cite_start]$

- **Difference and Ratio:** Techniques are also provided for $Z = X - Y$ and $Z = \frac{X}{Y}$ [cite: 211, 223].